

Data Loss Prevention — Guided Notes ANSWER KEY

Unit 4.4.2 | CompTIA Network+ N10-009 Obj. 4.5 | N10-008 4.5

For instructor use / distribute after students complete the notes.

Question 1

DLP addresses data in three states. Match each state to its description:

Data at _____ — stored on a device or server, protected primarily by encryption and access controls.

Data in _____ — being transmitted across a network, vulnerable to interception or misdirection.

Data in _____ — actively open or displayed, vulnerable to screen capture, clipboard copying, or printing.

Answer: Data at rest; data in motion; data in use.

Question 2

A DLP system uses two primary identification methods:

_____ inspection examines the actual file contents for patterns like SSN formats, credit card numbers, or keywords.

_____ awareness evaluates who is sending data, where it is going, and what application is being used — regardless of content.

Answer: Content inspection; context awareness.

Question 3

A guidance counselor emails a spreadsheet containing IEP records to a teacher's personal Gmail account. Which federal regulation does this most directly violate, and why?

Answer: FERPA (Family Educational Rights and Privacy Act). FERPA requires schools receiving federal funding to control access to and disclosure of student education records. Sending IEP records to a personal, non-district email address is an unauthorized disclosure of protected student records.

Question 4

Endpoint DLP can enforce controls directly on a user's device. List three specific endpoint DLP controls described in the lesson:

1. _____
2. _____
3. _____

Answer: Any three of: (1) USB drive blocking — prevents writes to removable media; (2) Clipboard controls — blocks copy/paste from protected apps to unprotected ones; (3) Print controls — prevents printing sensitive documents without authorization; (4) Screen capture controls — blocks Print Screen when protected apps are in focus.

Question 5

Network DLP requires _____ inspection to function effectively in modern environments. Without it, the DLP appliance can only see _____ blobs for most outbound traffic. What performance and privacy tradeoff does enabling this create?

Answer: SSL/TLS inspection; encrypted blobs. The appliance must decrypt, inspect, re-encrypt, and forward every HTTPS session — introducing processing overhead, latency, and privacy concerns because the organization can now read all encrypted employee communications.

Question 6

A _____ (CASB) sits between users and cloud services to enforce DLP policies on cloud-hosted data. Explain why on-premise network DLP is insufficient when teachers access Google Drive from home.

Answer: Cloud Access Security Broker (CASB). When a teacher accesses Google Drive from home, traffic travels between their home network and Google's servers — it never crosses the school's network boundary. The on-premise DLP appliance never sees the traffic, so it cannot inspect or block it. A CASB enforces policy regardless of the user's physical location.

Question 7

Data _____ assigns sensitivity labels such as Public, Internal, Confidential, and Restricted to data. DLP rules then reference these labels to apply appropriate controls. Why does this system often fail in practice, even when the technology works correctly?

Answer: Data classification. It fails because classification requires consistent, disciplined behavior from every user who creates or handles data — people must actually label things correctly. Manual classification relies on user training and compliance, which are inherently inconsistent. The technology enforces the label; the human applies it.

Question 8 — Real World Application

REAL WORLD: A user emails a sensitive spreadsheet as a password-protected ZIP file. The network DLP system allows it through without flagging it. A colleague complains that DLP is blocking legitimate encrypted emails to a vendor. Identify both failure modes and explain the tradeoff the security team faces when deciding how to handle encrypted outbound attachments.

Answer: First scenario: encryption bypass failure mode — content inspection cannot read inside a password-protected archive, so the sensitive data leaves undetected. Second scenario: false positive — blocking all encrypted attachments prevents legitimate secured communications. The tradeoff: allow encrypted attachments (risk data exfiltration via encryption bypass) vs. block all encrypted attachments (disrupt legitimate encrypted workflows). Most organizations choose a middle path: block unknown encrypted archives but allow organizationally-managed encrypted email.